

Sanjay Kumar Kureel, PhD

Independent Research Scholar

Cell Biology | Aging | Mechanobiology | Neurodegeneration

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My research focuses on the cellular and molecular mechanisms underlying aging, cellular senescence, and neurodegeneration. Integrating cell biology, mechanobiology, and translational neuroscience, I elucidate how mechanical and chemical signals regulate the age associated neurodegenerative disorders and identify biological mechanisms with therapeutic potential.

Professional Research Experience

- **Independent Research Scholar (Aug 2026-Present)**

United States

Research focus: Prepare review articles and develop research proposals on cellular senescence and neurodegeneration

- **Postdoctoral Research Fellow, (Feb 2025-July 2026)**

Department of Cellular and Integrative Physiology
Sam and Ann Institute for Longevity and Aging studies
UT Health San Antonio
San Antonio Texas

- **Postdoctoral Fellow, (Feb 2020- Feb 2025)**

Department of Biochemistry and Molecular Biology,

The University of Texas Medical Branch (UTMB) Galveston
Galveston, Texas

- **Research Assistant (March 2019-Sep 2019)**
Department of Chemical Engineering, Indian Institute of Technology Bombay, India.

Education

- **Ph.D. in Chemical Engineering [2014-2019]**
Indian Institute of Technology (IIT) Bombay, India.
Advisor: Prof. Abhijit Majumder
- **Master of Technology in Chemical Engineering [2006–2008]**
Indian Institute of Technology (IIT) Kharagpur, India.
Advisor: Prof. J.K. Basu
- **Bachelor of Technology in Chemical Engineering [1998-2002]**
Harcourt Butler Technological University (H.B.T.U.) Kanpur, India.

Research Interests

- Cellular and molecular mechanisms of aging, cellular senescence and age-related cognitive decline
- Mechanobiology of cognitive impairments, stem cell biology, and rejuvenation
- Translational neuroscience and biomarker discovery
- Develop methods to rejuvenate and regenerate tissues and organs
- Impact of pathological mechanical memory on Hippocampus functions

Experimental Expertise

Translational and Functional Assays

- Mouse handling, colony maintenance, survival surgery
- Behavioral assessments (treadmill, rotarod, grip strength, and inverted cling, and voluntary wheel running)
- Functional phenotyping of aging and disease models

Molecular Techniques

- Primary mammalian cell culture, stem cells, and established cancer cell lines
- Stem cell differentiation assays
- RT-qPCR; DNA/RNA extraction and purification
- Immunofluorescence and immunohistochemistry
- Flow cytometry
- Protein extraction and purification; ELISA; HiBiT reporter assays
- Gel electrophoresis and plasmid DNA recovery/transformation
- Confocal and super-resolution microscopy
- Traction force microscopy
- Engineering and culture of cells on tunable mechanical substrates, including polyacrylamide hydrogels and PDMS platforms

Computational & Data Analysis

- Statistical and transcriptomic analysis (RNA-Seq), data visualization using **R** and **Python**
- GraphPad Prism and imageJ/Fiji
- Scientific figure preparation using **BioRender** and **Adobe Illustrator**

Publications

Manuscripts in Preparation / Under Review

1. **Kureel, S.K.**, et al. *Enrichment of nuclear YAP1 rejuvenates senescent cells*. Manuscripts in Preparation.
2. **Kureel, S.K.**, Rasmussen, B. *Nicotinamide mononucleotide ameliorates progerin-associated defects and restores nuclear morphology in Hutchinson–Gilford progeria syndrome cells*. Manuscript Under Review.
3. **Kureel, S.K.**, et al. Low-frequency ultrasound attenuates amyloid- β and tau pathology while enhancing autophagy, synaptic function, and metabolic resilience in 3xTg Alzheimer's disease mice. Manuscripts in preparation.

Peer-Reviewed Publications.

4. **Kureel, S.K.**, Maroto, R., Aniqua, M., Powell, S., Singh, E., Margadant, F., et al. *Rejuvenation of senescent cells, in vitro and in vivo, by low-frequency ultrasound*. **Aging Cell**. 2025; 24(6): e70008.

5. **Kureel, S.K.**, Rasmussen, B.B. *Targeting ferroptosis to eliminate senescent cells: mechanisms and therapeutic potential*. **Aging and Disease**. 2025.
6. Has, C., **Kureel, S.K.** *Advances in microfluidic- and artificial intelligence-assisted design of liposomes for drug-delivery applications*. **Journal of Liposome Research**. Accepted.
7. Marchant, E.D., Singh, E., **Kureel, S.K.**, Blair, B., Kalenta, H., Von Ruff, Z.D., et al. *Low-frequency ultrasound reverses insulin resistance and diabetes-induced alterations in the muscle transcriptome of aged mice*. **American Journal of Physiology – Endocrinology and Metabolism**. 2025; 328(6): E899–E910.
8. **Kureel, S.K.**, Maroto, R., Davis, K., Sheetz, M.P. *Cellular mechanical memory: a potential tool for mesenchymal stem cell-based therapy*. **Stem Cell Research & Therapy**. 2025; 16(1):159.
9. **Kureel, S.K.**, Blair, B., Sheetz, M.P. *Recent advances in elimination strategies and rejuvenation targets of cellular senescence*. **Advanced Biology (Weinheim)**. 2024; 8(1): e2300461.
10. **Kureel, S.K.**, Sinha, S., Purkayastha, P., Barretto, S., Majumder, A. *Substrate stiffness controls the cell cycle of human mesenchymal stem cells via cellular traction*. **JOM**. 2022; 74(9): 3419–3427.
11. **Kureel, S.K.**, Mogha, P., Khadpekar, A., Kumar, V., Joshi, R., Das, S., et al. *Soft substrates preserve proliferative and adipogenic potential of mesenchymal stem cells by delaying senescence*. **Biology Open**. 2019; 8(4).
12. Mogha, P., Srivastava, A., Kumar, S., Das, S., **Kureel, S.K.**, Dwivedi, A., et al. *Hydrogel scaffolds with physiologically relevant elasticity promote proliferation of functional keratinocytes*. **RSC Advances**. 2019; 9(18): 10174–10183.

Book Chapters

1. Has, C., **Kureel, S.K.** *Adsorption equilibrium*. In: **Adsorption Dynamics: From Technologies to Environmental Solutions**. Cambridge Scholars Publishing; 2026.
2. Kureel, S. K., Namita. *Linking Mechanical Memory to Traumatic Brain Injury (TBI)-Induced Persistent Hippocampus Dysfunction*. In: **The Hippocampus - Architecture, Cognition, Plasticity and Dysfunction**. IntechOpen; 2026 (accepted for publication).

Patents

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- **Reversal of Senescence by Ultrasound Irradiation**
Inventors: M. Sheetz, S. K. Kureel, B. Rasmussen, R. Maroto
US Patent Application: 18/719,116

- **Reversal of Cellular Senescence by Low-Frequency Ultrasound Treatment**
Inventors: M. Sheetz, S.K. Kureel, F.M. Margadant
US Patent Application: 18/217,362

Academic and Scientific Appointments

- Postdoctoral Research Fellow, UT Health San Antonio, USA (Feb 2025-July 2026)
- Assistant Professor, Institute of Technology and Management (ITM), Gwalior, India (2011–2013)
- Assistant Professor, Anand Engineering College (AEC), Agra, India (2009–2011)

Honors and Awards

- **Best Poster Award**, Annual Symposium of Aging, UTMB Galveston, USA (2024)
- **Best Poster Presentation**, ComFlu International Conference, IISER Bhopal, India (2019)
- **Best Poster Presentation**, ChEmference '18, IIT Bombay, India
- **Best Poster Presentation**, International Congress of Cell Biology, NCCS Hyderabad, India (2018)
- **Best Poster Presentation**, Recent Advances in Chemical Biology, CEBS, Mumbai, India
- **Best Teacher Award**, Institute of Technology & Management, Gwalior, India (2013)

Invited Seminars and Presentations

- Mechanical Forces in Cellular Rejuvenation and Healthy Aging
Cellular & Integrative Physiology Symposium, UT Health San Antonio, USA (2026)
- Mechano-Regulation of Aging and Rejuvenation
Friday at Barshop Seminar Series, UT Health San Antonio, USA (2025)
- Low-Frequency Ultrasound Rejuvenation of Senescent Cells In Vitro and In Vivo
Translational Research in Aging & Metabolism, UTMB, USA (2022)
- Soft Substrate-Mediated Regulation of Stem Cell Function

Mechanobiology Institute, Singapore (2019)

Scientific Peer Review

Journals Peer Reviewer

- Stem Cell Research & Therapy
- Aging and Disease
- NPJ Aging
- Science Advances
- STAR Protocols
- Biogerontology
- European Journal of Medical Research

Research Mentoring and Leadership Experience

- Mentored research assistants in cell-based and animal studies, including training in experimental design, protocol development, data acquisition, and quantitative analysis (UTMB Galveston; 2020–2025).
- Supervised and mentored PhD rotation students, providing guidance on project design, experimental execution, and data interpretation (UTMB Galveston; 2020–2022).
- Mentored graduate and master's-level students during doctoral training, with a focus on research methodology, experimental rigor, and data analysis (IIT Bombay; 2014–2019).

Professional Memberships

- Tissue Engineering and Regenerative Medicine International Society (TERMIS)
- Biophysical Society (BPS)
- American Aging Association
- American Physiological Society (APS)
- International Cell Senescence Association (ICSA)

Editorial and Peer-Review Service

- Review Board Member, *Stem Cell Research & Therapy*
- Review Board Member, *Bio-Protocol*
- Managing editor: *Molecular and Cellular Biochemistry*
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